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Two-Grid Discontinuous Galerkin Method for Quasi-Linear Elliptic Problems

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Abstract In this paper, we consider the symmetric interior penalty discontinuous Galerkin (SIPG) method with piecewise polynomials of degree $r \geq 1$ for a class of quasi-linear elliptic problems in $\Omega \subset \mathbb{R}^2$. We propose a two-grid approximation for the SIPG method which can be thought of as a type of linearization of the nonlinear system using a solution from a coarse finite element space. With this technique, solving a quasi-linear elliptic problem on the fine finite element space is reduced into solving a linear problem on the fine finite element space and solving the quasi-linear elliptic problem on a coarse space. Convergence estimates in a broken H^1-norm are derived to justify the efficiency of the proposed two-grid algorithm. Numerical experiments are provided to confirm our theoretical findings. As a byproduct of the technique used in the analysis, we derive the optimal pointwise error estimates of the SIPG method for the quasi-linear elliptic problems in $\mathbb{R}^d$, $d = 2, 3$ and use it to establish the convergence of the two-grid method for problems in $\Omega \subset \mathbb{R}^3$.

Keywords Discontinuous Galerkin method · SIPG · Quasi-linear elliptic · Two-grid algorithm · Superconvergence

1 Introduction

The development and study of the discontinuous Galerkin (DG) methods have seen intensive research activities since they were proposed by Reed and Hill [37] for solving hyperbolic

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neutron transport equations The DG methods offer various advantages, among which are the ability to handle various element's types and shapes, irregular non-matching grids, and even locally varying polynomial order. We refer the reader to [2, 17, 24, 38] and the references therein for a historical review of various DG methods. In particular, considerable attention has been made to the development and analysis of the DG methods for elliptic problems. We refer the reader to [2, 4, 9–14, 16, 21, 26, 28–30, 32, 34, 35, 39] for the DG methods of linear elliptic problems, to [12, 22, 23, 25, 31, 36] for the DG methods of nonlinear elliptic problems.

Recently, Houston, Robson and Süli [25] considered the discontinuous hp-Galerkin methods for a family of quasi-linear elliptic problems which are of monotone type and obtained the *a priori* error estimate in a broken H^1-norm. Gudi and Pani [22] considered the discontinuous hp-Galerkin methods for a class of quasi-linear elliptic problems which are of nonmonotone type, and obtained the *a priori* error estimates in a broken H^1-norm and in the L^2-norm, which are optimal in h and suboptimal in p.

However, discontinuous Galerkin methods can potentially create algebraic equations with a huge number of degree of freedoms, and this is considered as one of the main hurdles in finding their solutions. To make matter worse, the boundary value problems in which the methods are applied to are inherently nonlinear. Thus the natural picture is solving numerically a system of nonlinear algebraic equations having very large dimensions. Typical strategy is to devise some fixed point type iterations or carrying out Newton's like iterations. Still, once either of the "linearization" is in place, the associated large dimensional linear algebra needs to be solved.

To this end, it is already clear that there is a dire need to address the difficulty. The appropriate setting is to derive an efficient algorithm that would solve the system in reasonably inexpensive fashion and yet maintain certain level of accepted accuracy. Our thesis is that for wide range of applications of discontinuous Galerkin methods, the two-grid algorithm is a viable approach. A premise in the two-grid algorithm is the understanding that nonlinear problems are easier to solve on a coarser mesh. On the other hand, many applications desire solutions that are resolved on the fine scale pertaining to the differential equations at hand. For this purpose, the two-grid algorithm uses the crude approximation on the coarse mesh to linearize the system to be solved on the fine mesh.

Within the continuous Galerkin framework, the two-grid finite element methods based on two finite element spaces on one coarse and one fine grid were first introduced by Xu [42–44] for the nonsymmetric linear and nonlinear elliptic problems. Later on, the two-grid methods were further investigated by many authors, for instance, Xu and Zhou [45] for eigenvalue problems, Axelsson and Layton [3] for nonlinear elliptic problems, Dawson, Wheeler and Woodward [18] for finite difference method for nonlinear parabolic equations, Utnes [40] for Navier-Stokes equations, Marion and Xu [33] for evolution equations, Wu and Allen [41] for mixed finite element method to solve coupled reaction-diffusion systems, and Bi and Ginting [5] for the finite volume element method for linear and nonlinear elliptic problems. These past works have given an indication on the viability of the two-grid methods as an efficient technique for solving nonlinear problems of various types.

To the best of our knowledge, the present work is the first attempt to apply and analyze the two-grid technique in the setting of the SIPG method. We propose to use this technique to solve the quasi-linear elliptic boundary value problems of nonmonotone type. The analysis presented in the main body of the paper is concentrated on problems in $\Omega \subset \mathbb{R}^2$. The discretization is based on one coarse and one fine discontinuous finite element spaces, V_H and V_h, respectively, living on a coarse grid of size H and a fine grid of size h and consisting of piecewise polynomials of degree $r \geq 1$. With this technique, solving a quasi-linear elliptic

problem on the fine space V_h is reduced into solving a linear problem on the fine space V_h and solving the quasi-linear elliptic problem on a coarse space V_H. This means that solving a quasi-linear elliptic problem is not much more difficult than solving one linear problem, since $\dim V_H \ll \dim V_h$ and the work for solving the quasi-linear problem on the coarse grid is relatively small. We show the convergence rate of the proposed two-grid algorithm in the broken H^1-norm. The convergence analysis of two-grid algorithm relies on the L^2-norm error estimate of the SIPG method for quasi-linear elliptic problems developed in [22].

As a byproduct of the technique used in the analysis, we derive the pointwise error estimates of the SIPG method for the quasi-linear elliptic problems in $\Omega \subset \mathbb{R}^d$, $d = 2, 3$. With the help of the error estimate in the L^2-norm, we get a superconvergence estimate in a broken H^1-norm between the SIPG solution and the so-called Ritz projection (see Appendix for a complete description). The pointwise estimates are then established by employing this superconvergence along with estimates in [14], and discrete Sobolev inequality. We then use this estimate to establish the convergence of the two-grid algorithm for solving SIPG method in $\Omega \subset \mathbb{R}^3$.

The remainder of this paper is organized as follows. We introduce notation and describe the SIPG method for the quasi-linear elliptic problems in Sect. 2. The two-grid algorithm of the SIPG method is proposed in Sect. 3, which is then followed by derivation of its convergence estimate in the broken H^1-norm. Sect. 4 provides numerical experiments to confirm our theoretical findings. In Appendix, we derive the pointwise error estimates for the SIPG method for the quasi-linear elliptic problems and use them to analyze the two-grid algorithm for solving SIPG method in $\Omega \subset \mathbb{R}^3$.

Throughout this paper, the letter C denotes a generic positive constant independent of the mesh parameter and may be different at different occurrences. Similarly, the notation $C(u)$ is used to denote a generic positive constant, signifying its dependence on u.

2 SIPG Method for Quasi-Linear Elliptic Problem

2.1 Preliminaries

Let Ω be a bounded polygonal domain in $\mathbb{R}^2$ with boundary $\partial\Omega$. Let $\mathcal{T}_h = \{K\}$ denote a shape-regular triangulation of the domain Ω, with K being a triangle or a quadrilateral with the diameter h_K and $h = \max\{h_K : K \in \mathcal{T}_h\}$. Denoting ρ_K the diameter of the inscribed circle to K, shape-regularity and quasi-uniformity mean respectively

$$h_K/\rho_K \le \varrho \quad \text{and} \quad h \le \varrho h_K, \quad \forall K \in \mathcal{T}_h, \tag{2.1}$$

for some constant $\varrho > 0$ which is independent of h and $K \in \mathcal{T}_h$.

Conformity of $\mathcal{T}_h$ is not a requirement for presentation of the results in this paper. In fact, we allow the mesh $\mathcal{T}_h$ to be *one-irregular*, i.e., each edge of any one element contains at most one hanging node. For simplicity, we assume this to be the midpoint of the corresponding edge. Details of this setting can be seen for example, in [27].

We use standard notation for the Sobolev spaces $W^{k,p}(D)$, $1 \le p \le \infty$, with norm $\|\cdot\|_{k,p,D}$ and seminorm $|\cdot|_{k,p,D}$, for polygonal Lipschitz domain $D \subset \mathbb{R}^2$ (see for example [8, 15]). When $p = 2$, we write $W^{k,2}(D) = H^k(D)$, and we omit writing index D whenever $D = \Omega$. Furthermore, for $k = 0$, we denote the norm by $\|\cdot\|$ which is the standard L^2-norm. We define as usual the space $H_0^1(\Omega) = \{v \in H^1(\Omega) | v = 0 \text{ on } \partial\Omega\}$. In what follows, the symbol $|\cdot|$ denotes either the length of a line segment or the area of a plane face, and

$(\cdot,\cdot)$ denote the $L^2(\Omega)$ inner product. For a positive integer k, we define the discontinuous Sobolev space on the triangulation $\mathcal{T}_h$ by,

$$H^k(\mathcal{T}_h) = \{v \in L^2(\Omega) : v|_K \in H^k(K),\ \forall K \in \mathcal{T}_h\},$$

equipped with the norm and the seminorm, respectively,

$$\|v\|_{k,h} = \left(\sum_{K\in\mathcal{T}_h} \|v\|_{k,K}^2\right)^{\frac{1}{2}}, \qquad |v|_{k,h} = \left(\sum_{K\in\mathcal{T}_h} |v|_{k,K}^2\right)^{\frac{1}{2}}.$$

We also assume that each element $K \in \mathcal{T}_h$ is an affine image of a fixed reference element $\widehat{K}$, i.e., $K = F_K(\widehat{K})$ for each element $K \in \mathcal{T}_h$, where $\widehat{K}$ is either the shape regular master triangle or rectangle in $\mathbb{R}^2$. For an integer $r \geq 1$, $\mathcal{P}_r(\widehat{K})$ denotes the set of polynomials of total degree r on the triangle $\widehat{K}$. When $\widehat{K}$ is the rectangle, $\mathcal{Q}_r(\widehat{K})$ denotes the set of all tensor-product polynomials on $\widehat{K}$ of degree r in each coordinate direction. Then, we introduce the discontinuous finite element space

$$V_h = \{v \in L^2(\Omega) : v|_K \cdot F_K \in \mathcal{R}_r(\widehat{K}),\ r \geq 1,\ \forall K \in \mathcal{T}_h\},$$

where $\mathcal{R}_r(\widehat{K})$ is either $\mathcal{P}_r(\widehat{K})$ or $\mathcal{Q}_r(\widehat{K})$.

Next, we define the average and jump operators that appear in the SIPG method. To this end, we denote by $\mathcal{E}_h$ the set of all open edges of all elements $K \in \mathcal{T}_h$, and $\mathcal{E}_h^0$ the set of all interior edges of $\mathcal{T}_h$. Let K^+ and K^- be two adjacent elements of $\mathcal{T}_h$ which share a common edge E, $E = \partial K^+ \cap \partial K^-$. Furthermore, let v and q be scalar- and vector-valued functions, respectively, that are smooth inside each element $K^\pm$. We designate $v^\pm$ and $q^\pm$ the traces of v and q on E taken from within the interior of $K^\pm$, respectively. The average and jump of v and q on E are respectively given by

$$\{v\} = \frac{1}{2}(v^+ + v^-), \qquad [v] = v^+\mathbf{n}_{K^+} + v^-\mathbf{n}_{K^-};$$
$$\{q\} = \frac{1}{2}(q^+ + q^-), \qquad [q] = q^+ \cdot \mathbf{n}_{K^+} + q^- \cdot \mathbf{n}_{K^-},$$

where $\mathbf{n}_{K^\pm}$ denote the unit outward normal vector of $\partial K^\pm$, respectively. On a boundary face $E \subset \partial\Omega$, we set $\{v\} = v$, $\{q\} = q$ and $[v] = v\mathbf{n}$ with $\mathbf{n}$ denoting the unit outward normal vector on the boundary $\partial\Omega$. It is clear that the jump $[v]$ of the scalar function v is a vector parallel to the normal, and the jump $[q]$ of the vector function q is a scalar quantity. The advantage of these definitions is that they do not depend on assigning an ordering to the elements $K^\pm$. We equip the space $H^1(\mathcal{T}_h)$ with the following broken H^1-norm defined by

$$|||v||| = \left(|v|_{1,h}^2 + \sum_{E\in\mathcal{E}_h} h_E^{-1}\|[v]\|_{0,E}^2\right)^{\frac{1}{2}}.$$

We will need the following trace inequality (see for example [1]).

Lemma 1 *Let E be an edge of $K \in \mathcal{T}_h$. There exists a positive constant C independent of $v \in H^1(K)$ such that*

$$\|v\|_{0,E}^2 \leq C(h_E^{-1}\|v\|_{0,K}^2 + h_E|v|_{1,K}^2).$$

2.2 SIPG Method for Quasi-Linear Elliptic Problem

The benchmark problem that we consider in this paper is a second-order quasi-linear elliptic problem

$$\begin{cases} -\nabla \cdot (a(u)\nabla u) = f(x), & \text{in } \Omega, \\ u(x) = 0, & \text{on } \partial\Omega. \end{cases} \tag{2.2}$$

We assume $0 < \alpha \le a(z) \le M$, $z \in \mathbb{R}$, for some positive constants α, M, and $a(z) \in C_b^2(\mathbb{R})$, where $C_b^2(\mathbb{R})$ is the space of twice continuously differentiable functions on $\mathbb{R}$ whose first and second order derivatives are bounded (see [19] and [22] for similar assumptions). The existence of unique weak solution of (2.2) in $C^{2+\delta}(\Omega)$, with $\delta \in (0, 1)$, is guaranteed when domain's boundary is smooth and $f \in C^{\delta}(\Omega)$, cf. e.g., [20]. However, because we assume the domain Ω is polygonal, we do not expect the solution of (2.2) to enjoy such regularity. Following [22], we assume that problem (2.2) has a solution $u \in H^2(\Omega) \cap W^{1,\infty}(\Omega)$.

For ω, u, and $v \in H^2(\mathcal{T}_h)$, we define the form $A(\omega; u, v)$, which is linear in u, v for fixed ω, by

$$\begin{aligned} A(\omega; u, v) = &\sum_{K \in \mathcal{T}_h} \int_K a(\omega)\nabla u \cdot \nabla v \mathrm{d}x - \sum_{E \in \mathcal{E}_h} \int_E \{a(\omega)\nabla u\} \cdot [v] \mathrm{d}s \\ &- \sum_{E \in \mathcal{E}_h} \int_E \{a(\omega)\nabla v\} \cdot [u] \mathrm{d}s + \gamma \sum_{E \in \mathcal{E}_h} h_E^{-1} \int_E [u] \cdot [v] \mathrm{d}s, \end{aligned} \tag{2.3}$$

where γ is a penalty parameter. The weak formulation of (2.2) suitable for the SIPG method is to find $u \in H^2(\mathcal{T}_h)$ such that

$$A(u; u, v) = (f, v), \quad \forall v \in H^2(\mathcal{T}_h). \tag{2.4}$$

The following lemma establishes the equivalence of (2.2) and (2.4) (see [22] for detail).

Lemma 2 *If the solution u of (2.2) is in $H^2(\Omega)$, then u satisfies (2.4). Conversely, if $u \in H^1(\Omega) \cap H^2(\mathcal{T}_h)$ is a solution of (2.4), then u satisfies (2.2) weakly.*

The SIPG approximation of u is: find $u_h \in V_h$ such that

$$A(u_h; u_h, v_h) = (f, v_h), \quad \forall v_h \in V_h. \tag{2.5}$$

The existence of discrete solution (2.5) was shown in [22] for sufficiently small h using Brouwer's fixed point theorem. The uniqueness can be established if the discrete solution map is Lipschitz continuous. Next, we state the a priori error estimates for solution of (2.5) in the broken H^1-norm and L^2-norm, which have been derived in [22].

Lemma 3 *Let $u \in W^{1,\infty}(\Omega) \cap H^{r+1}(\Omega)$, $r \ge 1$, and $u_h \in V_h$ be the solutions of (2.4) and (2.5), respectively. For sufficiently small h, there exists a positive constant C independent of h such that*

$$|||u - u_h||| \le C(u)h^r, \qquad \|u - u_h\| \le C(u)h^{r+1}.$$

Lemma 4 *Let $u \in W^{1,\infty}(\Omega) \cap H^{r+1}(\Omega), r \geq 1$, and $u_h \in V_h$ be the solutions of* (2.4) *and* (2.5), *respectively. For sufficiently small h, we have*

$$\|u - u_h\|_{0,\infty} \leq C(u)h^r.$$

Proof With $\chi \in V_h$, using approximation property and the inverse estimate, $|v_h|_{0,\infty,K} \leq Ch_K^{-1}|v_h|_{0,K}$, and Lemma 3, we get

$$\begin{aligned}\|u - u_h\|_{0,\infty} &\leq \|u - \chi\|_{0,\infty} + \|\chi - u_h\|_{0,\infty} \\ &\leq Ch^r\|u\|_{r+1} + Ch^{-1}\|\chi - u_h\| \\ &\leq Ch^r\|u\|_{r+1} + Ch^{-1}(\|\chi - u\| + \|u - u_h\|) \\ &\leq C(u)h^r, \end{aligned} \tag{2.6}$$

which completes the proof. □

The following lemma gives the stability on the discrete solution u_h in the $W^{1,\infty}$-seminorm for the quasi-linear elliptic problem which holds true for $r \geq 1$ and will be used in the convergence analysis of the two-grid method.

Lemma 5 *Let $u \in W^{1,\infty}(\Omega) \cap H^2(\Omega)$ and $u_h \in V_h$ be the solutions of* (2.4) *and* (2.5), *respectively. Then $|u_h|_{1,\infty} \leq C(u)$.*

Proof Using the triangle inequality, we write

$$|u_h|_{1,\infty} \leq |u_h - \chi|_{1,\infty} + |\chi|_{1,\infty}, \tag{2.7}$$

where χ is the approximation of u. In view of the approximation property [15]

$$\|v - \chi\|_{k,\infty,K} \leq Ch^{m-k-2/p}|v|_{m,p,K}, \tag{2.8}$$

with $k = 1$, $m = 2$, and $p = 2$, we have

$$|\chi|_{1,\infty} \leq |u - \chi|_{1,\infty} + |u|_{1,\infty} \leq C\|u\|_2 + |u|_{1,\infty}. \tag{2.9}$$

Using inverse estimate $|v_h|_{1,\infty,K} \leq Ch_K^{-1}|v_h|_{1,K}$, Lemma 3, and approximation property $\|v - \chi\|_{1,K} \leq Ch_K|v|_{2,K}$, we get

$$|u_h - \chi|_{1,\infty} \leq Ch^{-1}|u_h - \chi|_{1,h} \leq Ch^{-1}(|u_h - u|_{1,h} + |u - \chi|_{1,h}) \leq C(u). \tag{2.10}$$

Combining (2.7), (2.9) with (2.10) completes the proof. □

Finally, we state the coercivity of $A(\omega; \cdot, \cdot)$ which holds true for sufficiently large γ. We refer to [14] for its proof.

Lemma 6 *There exist two positive constants α_0 and γ_0 such that for $\gamma \geq \gamma_0$*

$$\alpha_0|||v_h|||^2 \leq A(\omega; v_h, v_h), \quad \forall \omega \in H^1(\mathcal{T}_h),\ \forall v_h \in V_h.$$

A crucial task in carrying out the analysis stems from the nonlinearity of elliptic coefficient. The recurring feature is estimation of $(a(u)-a(z))$ in various integral formulation. A common approach to estimate this difference is to employ the mean value theorem

$$a(u)-a(z)=(u-v)\tilde{a}_u(z), \tag{2.11}$$

where $\tilde{a}_u(z)=\int_0^1 a'(u+t(z-u))\mathrm{d}t$. In addition, we also impose $a(z)\in C_b^1(\mathbb{R})$. The next three lemmas expound various estimates associated with this nonlinear elliptic coefficient.

Lemma 7 *If $a(z)\in C_b^1(\mathbb{R})$, then*

$$\left|\sum_{K\in\mathcal{T}_h}\int_K (a(u)-a(z))\nabla w_h\cdot\nabla v_h\mathrm{d}x\right|\le C|w_h|_{1,\infty}\,\|u-z\|\,|v_h|_{1,h},$$

for w_h and v_h elements of V_h.

Proof The estimate is established by applying (2.11) and Cauchy-Schwarz inequality. □

Lemma 8 *If $a(z)\in C_b^1(\mathbb{R})$, then*

$$\left|\sum_{E\in\mathcal{E}_h}\int_E \{(a(u)-a(z))\nabla w_h\}\cdot[v_h]\mathrm{d}s\right|\le C|w_h|_{1,\infty}\left(\sum_{K\in\mathcal{T}_h}\sum_{e\in\partial K}h_e\left\|u-z\right\|_{0,e}^2\right)^{\frac{1}{2}}|||v_h|||,$$

for w_h and v_h elements of V_h.

Proof Using (2.11), Cauchy-Schwarz inequality, and definition of $|||\cdot|||$, we get

$$\begin{aligned}
&\left|\sum_{E\in\mathcal{E}_h}\int_E \{(a(u)-a(z))\nabla w_h\}\cdot[v_h]\mathrm{d}s\right|\\
&\quad=\left|\sum_{E\in\mathcal{E}_h}\int_E \{\tilde{a}_u(z)(u-z)\nabla w_h\}\cdot[v_h]\mathrm{d}s\right|\\
&\quad\le\sum_{E\in\mathcal{E}_h}\left(h_E^{\frac{1}{2}}\left\|\{\tilde{a}_u(z)(u-z)\nabla w_h\}\right\|_{0,E}\cdot h_E^{-\frac{1}{2}}\left\|[v_h]\right\|_{0,E}\right)\\
&\quad\le\left(\sum_{E\in\mathcal{E}_h}h_E\left\|\{\tilde{a}_u(z)(u-z)\nabla w_h\}\right\|_{0,E}^2\right)^{\frac{1}{2}}\cdot\left(\sum_{E\in\mathcal{E}_h}h_E^{-1}\left\|[v_h]\right\|_{0,E}^2\right)^{\frac{1}{2}}\\
&\quad\le\left(\sum_{K\in\mathcal{T}_h}\sum_{e\in\partial K}h_e\left\|\tilde{a}_u(z)(u-z)\nabla w_h\right\|_{0,e}^2\right)^{\frac{1}{2}}\cdot|||v_h|||\\
&\quad\le C|w_h|_{1,\infty}\left(\sum_{K\in\mathcal{T}_h}\sum_{e\in\partial K}h_e\left\|u-z\right\|_{0,e}^2\right)^{\frac{1}{2}}\cdot|||v_h|||.
\end{aligned}$$

□

Lemma 9 *If $a(z) \in C_b^1(\mathbb{R})$, then*

$$\left|\sum_{E\in\mathcal{E}_h}\int_E \{(a(u)-a(z))\nabla v_h\}\cdot[w_h]\mathrm{d}s\right|$$
$$\le C\|u-z\|_{0,\infty}\left(\sum_{K\in\mathcal{T}_h}\sum_{e\in\partial K}\|\nabla v_h\|_{0,e}^2\right)^{\frac{1}{2}}\left(\sum_{K\in\mathcal{T}_h}\sum_{e\in\partial K}\|u-w_h\|_{0,e}^2\right)^{\frac{1}{2}},$$

for w_h and v_h elements of V_h.

Proof We use the fact that $[u]_E = 0$ for $E \in \mathcal{E}_h^0$ and $u|_E = 0$ for $E \subset \partial\Omega$, apply (2.11) and Cauchy-Schwarz inequality yielding

$$\left|\sum_{E\in\mathcal{E}_h}\int_E \{(a(u)-a(z))\nabla v_h\}\cdot[w_h]\mathrm{d}s\right|$$
$$=\left|\sum_{E\in\mathcal{E}_h}\int_E \{\tilde{a}_u(z)(u-z)\nabla v_h\}\cdot[u-w_h]\mathrm{d}s\right|$$
$$\le\sum_{E\in\mathcal{E}_h}\left(\|\{\tilde{a}_u(z)(u-z)\nabla v_h\}\|_{0,E}\cdot\|[u-w_h]\|_{0,E}\right)$$
$$\le\left(\sum_{E\in\mathcal{E}_h}\|\{\tilde{a}_u(z)(u-z)\nabla v_h\}\|_{0,E}^2\right)^{\frac{1}{2}}\cdot\left(\sum_{E\in\mathcal{E}_h}\|[u-w_h]\|_{0,E}^2\right)^{\frac{1}{2}}$$
$$\le\left(\sum_{K\in\mathcal{T}_h}\sum_{e\in\partial K}\|\tilde{a}_u(z)(u-z)\nabla v_h\|_{0,e}^2\right)^{\frac{1}{2}}\left(\sum_{K\in\mathcal{T}_h}\sum_{e\in\partial K}\|u-w_h\|_{0,e}^2\right)^{\frac{1}{2}}$$
$$\le C\|u-z\|_{0,\infty}\left(\sum_{K\in\mathcal{T}_h}\sum_{e\in\partial K}\|\nabla v_h\|_{0,e}^2\right)^{\frac{1}{2}}\left(\sum_{K\in\mathcal{T}_h}\sum_{e\in\partial K}\|u-w_h\|_{0,e}^2\right)^{\frac{1}{2}}.$$

□

3 Two-Grid Algorithm of SIPG Method

In this section, we present a two-grid algorithm of the SIPG method for the second-order quasi-linear elliptic problems (2.2). The basic mechanism in this algorithm is construction of two shape-regular subdivisions of Ω, which we denote by $\mathcal{T}_H$ and $\mathcal{T}_h$, with different mesh sizes H and $h(h \ll H)$. This can be accomplished by successively refining the coarse mesh $\mathcal{T}_H$ to obtain the fine mesh $\mathcal{T}_h$. The corresponding finite element spaces are V_H and V_h, which will be called coarse and fine space, respectively. We note that by such a construction we have $V_H \subset V_h$.

In this algorithm, we use the SIPG method to solve the quasi-linear elliptic problem on a coarse space V_H, and obtain a rough approximation $u_H \in V_H$, and use it to linearize the corresponding system on the fine space V_h, and solve the resulting linearized problem to get $u^h \in V_h$. This means that solving a quasi-linear equation is not much more difficult that

Algorithm 1 Two-grid SIPG Method

Find $u_H \in V_H$ such that $A(u_H; u_H, v_H) = (f, v_H), \forall v_H \in V_H$.
Find $u^h \in V_h$ such that $A(u_H; u^h, v_h) = (f, v_h), \forall v_h \in V_h$.

solving one linear equation, since $\dim V_H \ll \dim V_h$ and the work for solving u_H is relatively small. The following algorithm is the precise statement of this technique for solving (2.2).

Next we state and prove convergence estimates in the broken H^1-norm for the two-grid algorithm of the SIPG method under the assumption $u \in W^{1,\infty}(\Omega) \cap H^{r+1}(\Omega), r \geq 1$.

Theorem 1 *Assume that $u \in W^{1,\infty}(\Omega) \cap H^{r+1}(\Omega), r \geq 1$, $u_h \in V_h$ and $u^h \in V_h$ are the solutions obtained by* (2.4), (2.5) *and Algorithm* 1, *respectively. For $H \ll 1$, we have*

$$|||u_h - u^h||| \leq C(u)H^{r+1}; \tag{3.1}$$

$$|||u - u^h||| \leq C(u)(h^r + H^{r+1}). \tag{3.2}$$

Proof It follows from Algorithm 1 and (2.5) that

$$A(u_H; u^h, v_h) = (f, v_h) = A(u_h; u_h, v_h),$$

and thus using definition of $A(\cdot; \cdot, \cdot)$, we get

$$\begin{aligned} A(u_H; u^h - u_h, v_h) &= A(u_H; u^h, v_h) - A(u_H; u_h, v_h) \\ &= A(u_h; u_h, v_h) - A(u_H; u_h, v_h) \\ &= S_1 + S_2 + S_3, \end{aligned} \tag{3.3}$$

where

$$S_1 = \sum_{K \in \mathcal{T}_h} \int_K (a(u_h) - a(u_H))\nabla u_h \cdot \nabla v_h \mathrm{d}x,$$

$$S_2 = -\sum_{E \in \mathcal{E}_h} \int_E \{(a(u_h) - a(u_H))\nabla u_h\} \cdot [v_h] \mathrm{d}s,$$

$$S_3 = -\sum_{E \in \mathcal{E}_h} \int_E \{(a(u_h) - a(u_H))\nabla v_h\} \cdot [u_h] \mathrm{d}s.$$

In order to get the convergence property in the broken H^1-norm, we estimate terms on the right-hand side of (3.3). We write

$$\begin{aligned} S_1 &= \sum_{K \in \mathcal{T}_h} \int_K (a(u_h) - a(u))\nabla u_h \cdot \nabla v_h \mathrm{d}x + \sum_{K \in \mathcal{T}_h} \int_K (a(u) - a(u_H))\nabla u_h \cdot \nabla v_h \mathrm{d}x \\ &= S_{11} + S_{12}. \end{aligned} \tag{3.4}$$

To bound S_{11} and S_{12}, we use Lemma 7 with $z = u_h$ and $z = u_H$, respectively, $w_h = u_h$, Lemma 5 and Lemma 3

$$\begin{aligned} |S_1| &\leq C|u_h|_{1,\infty}(\|u - u_h\| + \|u - u_H\|)|v_h|_{1,h} \\ &\leq C(u)(\|u - u_h\| + \|u - u_H\|)|v_h|_{1,h} \end{aligned}$$

$$\begin{aligned}
&\le C(u)(h^{r+1}+H^{r+1})|v_h|_{1,h}\\
&\le C(u)H^{r+1}|||v_h|||.
\end{aligned} \tag{3.5}$$

We also write

$$\begin{aligned}
S_2 &= \sum_{E\in\mathcal{E}_h}\int_E \{(a(u)-a(u_h))\nabla u_h\}\cdot[v_h]\mathrm{d}s + \sum_{E\in\mathcal{E}_h}\int_E \{(a(u_H)-a(u))\nabla u_h\}\cdot[v_h]\mathrm{d}s\\
&= S_{21}+S_{22}.
\end{aligned} \tag{3.6}$$

Estimates for these two terms are established by using Lemma 8 with $z=u_h$ and $z=u_H$, respectively, $w_h=u_h$, and by applying Lemma 1, Lemma 5 and Lemma 3 to yield

$$\begin{aligned}
|S_2| &\le C|u_h|_{1,\infty}\left(\sum_{K\in\mathcal{T}_h}\sum_{e\in\partial K} h_e\|u-u_h\|_{0,e}^2\right)^{\frac{1}{2}}|||v_h|||\\
&\quad + C|u_h|_{1,\infty}\left(\sum_{K\in\mathcal{T}_h}\sum_{e\in\partial K} h_e\|u-u_H\|_{0,e}^2\right)^{\frac{1}{2}}|||v_h|||\\
&\le C(u)(h^{r+1}+H^{r+1})|||v_h|||\\
&\le C(u)H^{r+1}|||v_h|||.
\end{aligned} \tag{3.7}$$

Finally, we write for S_3,

$$\begin{aligned}
S_3 &= \sum_{E\in\mathcal{E}_h}\int_E \{(a(u)-a(u_h))\nabla v_h\}\cdot[u_h]\mathrm{d}s + \sum_{E\in\mathcal{E}_h}\int_E \{(a(u_H)-a(u))\nabla v_h\}\cdot[u_h]\mathrm{d}s\\
&= S_{31}+S_{32}.
\end{aligned} \tag{3.8}$$

Using Lemma 9 with $z=u_h$ and $w_h=u_h$, and applying Lemma 1, Lemma 4, Lemma 3 and (2.1), we have

$$\begin{aligned}
|S_{31}| &\le \|u-u_h\|_{0,\infty}\left(\sum_{K\in\mathcal{T}_h}\sum_{e\in\partial K}\|\nabla v_h\|_{0,e}^2\right)^{\frac{1}{2}}\left(\sum_{K\in\mathcal{T}_h}\sum_{e\in\partial K}\|u-u_h\|_{0,e}^2\right)^{\frac{1}{2}}\\
&\le C(u)h^r\left(\sum_{K\in\mathcal{T}_h} h_K^{-1}|v_h|_{1,K}^2\right)^{\frac{1}{2}}\left(\sum_{K\in\mathcal{T}_h} h_K^{-1}\|u-u_h\|_{0,K}^2 + h_K|u-u_h|_{1,K}^2\right)^{\frac{1}{2}}\\
&\le C(u)h^{2r}|||v_h|||.
\end{aligned} \tag{3.9}$$

Similarly $|S_{32}|\le C(u)H^{2r}|||v_h|||$, and therefore

$$|S_3|\le C(u)H^{2r}|||v_h|||. \tag{3.10}$$

Setting $v_h=u^h-u_h$ in (3.3) and using Lemma 6, (3.5), (3.7) and (3.10) yield (3.1). The estimate (3.2) is obtained by combining (3.1) with Lemma 3. □

Remark 1 Theorem 1 suggests that optimal rate of convergence in the two-grid SIPG method using discontinuous piecewise linear finite element space ($r = 1$) can be achieved by employing $H = \mathbf{O}(\sqrt{h})$.

4 Numerical Experiments

In this section we present numerical calculations pertaining to the theoretical finding presented in Sect. 3. Our focus is to illustrate the superconvergence of the two-grid discontinuous Galerkin solution to the standard discontinuous Galerkin finite element solution on the fine grid with respect to coarse mesh parameter (see (3.1) in Theorem 1). Moreover, it is also interesting to see that when the coarse mesh size is appropriately related to the fine mesh size, namely, $H = \mathbf{O}(\sqrt{h})$ for piecewise linear element, then (3.2) of Theorem 1 guarantees the optimal convergence of the two-grid discontinuous Galerkin finite element solution.

The examples we show here use linear approximation on triangles mesh. In practice, the solution of the SIPG method can be expressed as

$$u_h(x) = \sum_{K \in \mathcal{T}_h} \sum_{j=1}^{3} U_{K,j} \phi_{K,j}(x),$$

where $\phi_{K,j}$ is linear in K. Substituting this representation into the SIPG formulation yields a nonlinear system of algebraic equations $A(U)U = F$, where U is a vector whose entries are $U_{K,j}$s and F is a vector containing discretization of the right hand side (f, v_h) of (2.5). We have used a fixed point type iteration to solve this system, namely to write it as a linearization of the form $A(U^{(k-1)})U^{(k)} = F$, where $U^{(k-1)}$ is the previous iterate vector. This gives a linear system governing the next iterate vector $U^{(k)}$, which is solved by the GMRES method. The fixed point iteration is continued until a tolerance of $\|A(U^{(k-1)})U^{(k)} - F\|_{l_\infty} < 10^{-8}$ is reached. We note that in this setting, solving u^h in the two-grid SIPG method can be regarded as a one time fixed point iteration with $U^{(0)}$ chosen as the components of u_H. Of course, u_H in this case is still obtained from the fixed point iteration mentioned above.

In all examples, the problems are posed in domain $\Omega = [0, 1] \times [0, 1]$. The domain is discretized into N numbers of rectangle in each direction and then each rectangle is divided into two triangles, resulting in a mesh with size $\sqrt{2}/N$. For the superconvergence of two-grid solution, we compute u_H on the coarse finite element space V_H, with $H = \sqrt{2}/N$, $N = 4, 8, 16, 32, 64$, and u^h on the fine finite element space V_h, with $h = \sqrt{2}/256$. To see the optimal convergence of the two-grid solution in the broken norm, we compute u_H with $H = \sqrt{2}/N$, $N = 4, 6, 8, 10, 12, 14$ and u^h, with $h = \sqrt{2}/N$, $N = 16, 36, 64, 100, 144, 196$, thereby maintaining relation $H = c\sqrt{h}$. In all of the figures below, a least-squares fit is conducted for each of the associated errors.

The first problem uses

$$a(u) = 1 - 0.9\sin(4\pi u) \quad \text{and} \quad u(x_1, x_2) = (x_1 - x_1^2)(x_2 - x_2^2), \tag{4.1}$$

where $f(x)$ is chosen such that $f(x) = -\nabla \cdot (a(u)\nabla u)$ for this pair of equations. Figure 1 shows the log-log plot of $|||u_h - u^h|||$ against H for several values of penalty parameter γ. The rate of convergence of this deviation ranges from 1.91 to 1.97. This confirms the first part of Theorem 1. Furthermore, Fig. 2 shows convergence of $|||u - u^h|||$ with configuration $H = c\sqrt{h}$ mentioned above. The rate of convergence varies between 0.98 to 1.03, which

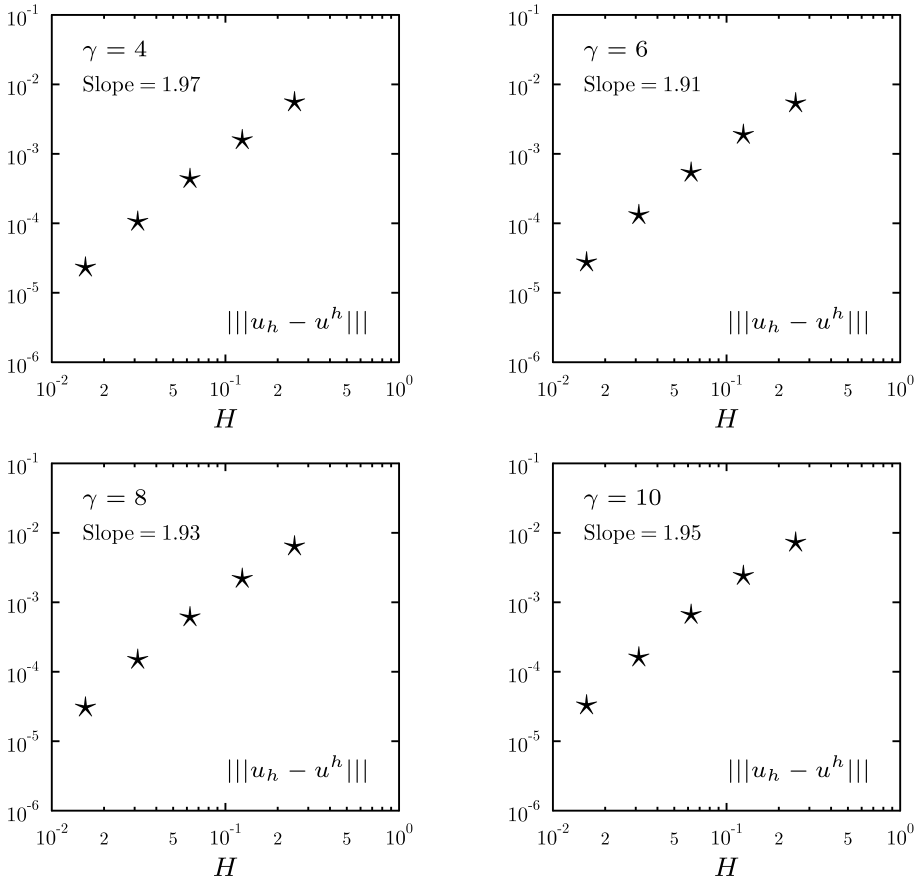

Fig. 1 Superconvergence of two-grid SIPG solution of (2.2) with elliptic coefficient and true solution as in (4.1)

has been predicted by the second part of Theorem 1. It is also worth mentioning that these convergence behaviors are maintained for different values of penalty parameter γ.

For the second problem we consider

$$a(u) = 1 - 0.9\sin(20\pi u) \quad \text{and} \quad u(x_1, x_2) = x_1^3 \ln(x_1) x_2^3 \ln(x_2), \tag{4.2}$$

where as before, $f(x)$ is chosen such that $f(x) = -\nabla \cdot (a(u)\nabla u)$ for this pair of equations. The choice of this example reflects a condition which is less smooth compared to the first one. The corresponding convergence behaviors are shown in Figs. 3 and 4. As expected, the convergence rates are slightly worse than in the first example. Nevertheless, their overall behaviors are still within the prediction of Theorem 1.

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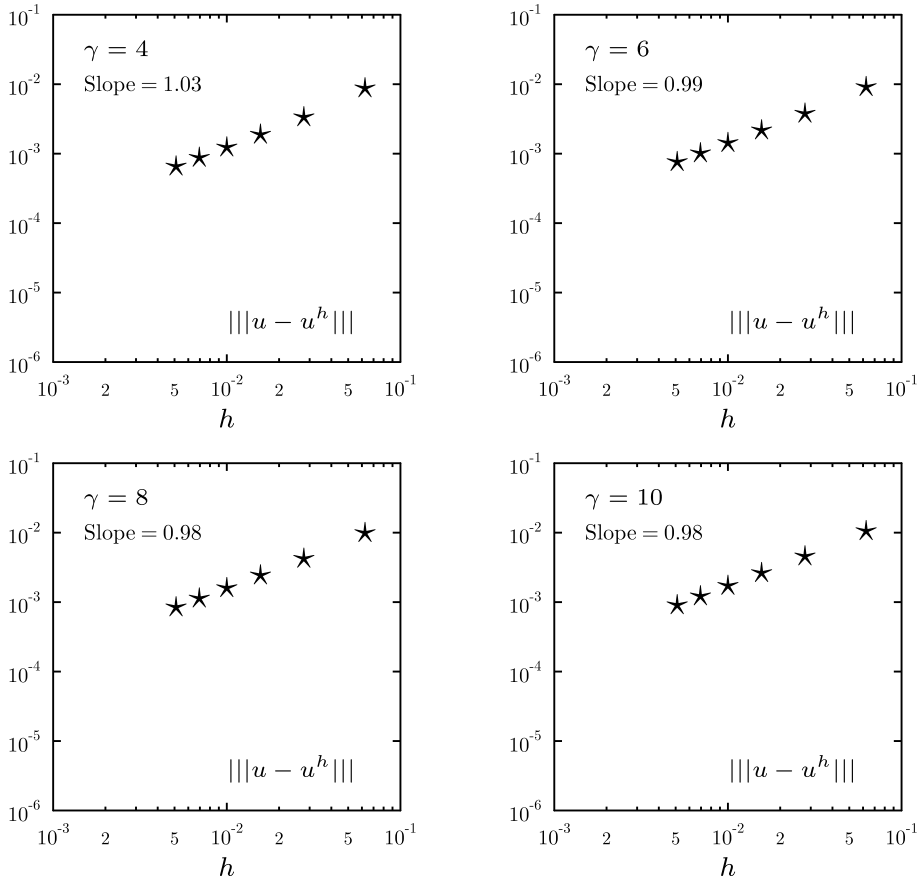

Fig. 2 Convergence of two-grid SIPG solution of (2.2) with elliptic coefficient and true solution as in (4.1)

Appendix: Pointwise Error Estimates for SIPG Method and Two-Grid Algorithm for Problems in 3D

In this appendix, we derive the pointwise error estimates of the SIPG method for the quasi-linear elliptic problems. We intend to establish the estimates for problems in $\Omega \subset \mathbb{R}^d$, $d = 2, 3$. As a comparison, local pointwise estimates have been established by Kanschat and Rannacher [28] and Chen and Chen [14] for SIPG method applied to the second-order linear elliptic problems. After establishing these estimates, we proceed with the convergence analysis of two-grid SIPG method applied to problem in three-dimensional space.

Before embarking on the details of the analysis, we need to extend some geometrical definitions described in Sect. 2.1 that will make the SIPG method amenable for solving boundary value problems in three dimensional space setting. We think of Ω as a bounded open polyhedral domain, and $\mathcal{T}_h = \{K\}$ a subdivision of Ω into disjoint open element domain K such that $\overline{\Omega} = \bigcup_{K \in \mathcal{T}_h} \overline{K}$, where $\mathcal{T}_h$ is regular or *one-irregular*, i.e., each face of K contains at most one hanging node, the barycenter of the face. Details of this setting can be seen for example, in [27]. We assume that $\mathcal{T}_h$ is shape-regular and quasi-uniform. The set $\mathcal{E}_h$ is thought of to contain open faces of all element $K \in \mathcal{T}_h$.

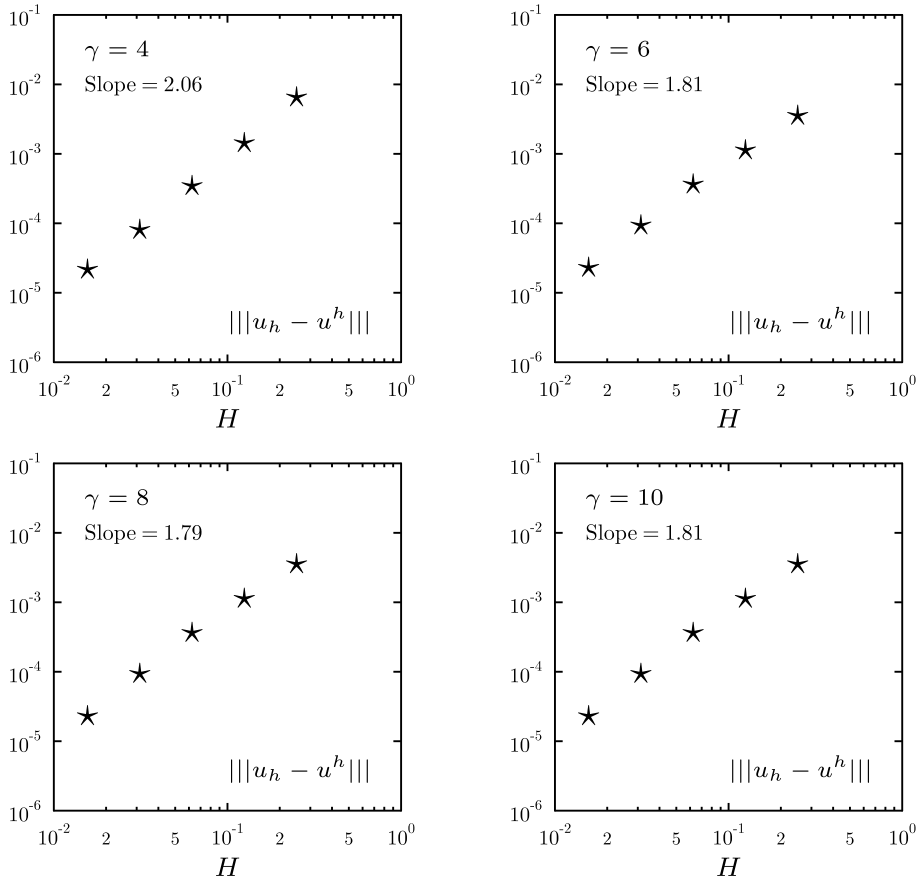

Fig. 3 Superconvergence of two-grid SIPG solution of (2.2) with elliptic coefficient and true solution as in (4.2)

At this stage, we emphasize on the facts that most of the mathematical tools in Sect. 2.1 following the geometrical definitions remain true for $\Omega \subset \mathbb{R}^3$. We can extend the formulation of SIPG in Sect. 2.2 to problems in $\Omega \subset \mathbb{R}^3$.

A. Pointwise Error Estimates for SIPG Method

The first step toward establishing the pointwise error estimates requires comparing the SIPG solution against the elliptic projection $R_h : H^2(\Omega) \cap H_0^1(\Omega) \to V_h$ defined by

$$A(u; R_h u, v_h) = A(u; u, v_h), \quad \forall v_h \in V_h. \tag{A.1}$$

In particular, we establish a superconvergence for $(u_h - R_h u)$ in the broken H^1-norm. Notice that for fixed $u \in H^2(\Omega) \cap H_0^1(\Omega)$, (A.1) is a linear variational problem. Approximation properties of R_h are stated in the following two lemmas (see [2] for the proof of Lemma A.1 below).

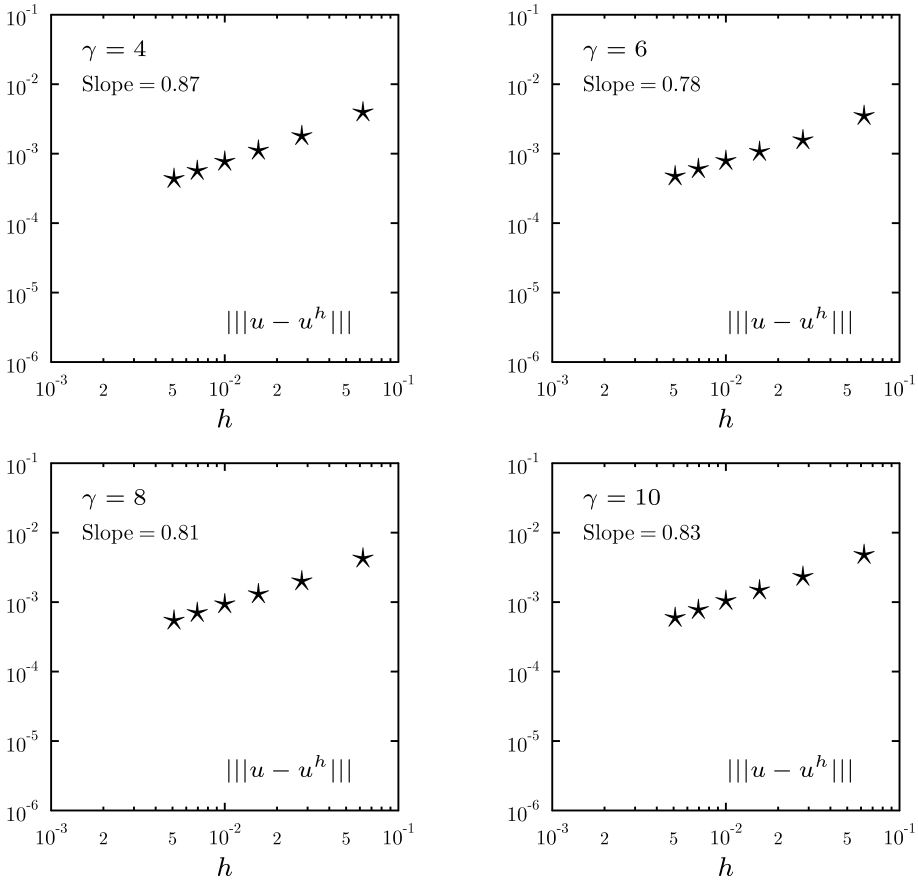

Fig. 4 Convergence of two-grid SIPG solution of (2.2) with elliptic coefficient and true solution as in (4.2)

Lemma A.1 *There exists a positive constant C independent of h such that*

$$\|u - R_h u\| + h|||u - R_h u||| \le C(u)h^{r+1}, \tag{A.2}$$

for $u \in H^{r+1}(\Omega) \cap H_0^1(\Omega)$ and $R_h u$ defined by (A.1).

The following pointwise error estimates for the linear problem (A.1) have been developed in [14].

Lemma A.2 *Let $u \in W^{r+1,\infty}(\Omega) \cap H_0^1(\Omega)$ and $R_h u \in V_h$ be defined by* (A.1). *There exists a positive constant C independent of h such that*

$$\|u - R_h u\|_{0,\infty} \le C(u)h^{r+1}|\ln h|^{\bar{r}}, \tag{A.3}$$

and

$$\|\nabla(u - R_h u)\|_{0,\infty} \le C(u)h^{r}, \tag{A.4}$$

where $\bar{r} = 0$ *if* $r > 1$ *and* $\bar{r} = 1$ *if* $r = 1$.

Using (A.4), we can easily get the following lemma by using the triangle inequality.

Lemma A.3 *For* $u \in W^{2,\infty}(\Omega) \cap H_0^1(\Omega)$, *we have* $|R_h u|_{1,\infty} \le C(u)$.

Now we are in position to prove the superconvergence of $|||u_h - R_h u|||$.

Theorem A.1 *Let* $u \in W^{r+1,\infty}(\Omega)$ *and* $u_h \in V_h$ *be the solution of* (2.4) *and* (2.5), *respectively. For sufficiently small* h, *there exists a positive constant* C *independent of* h *such that*

$$|||u_h - R_h u||| \le C(u) h^{r+1}. \tag{A.5}$$

Proof In view of (2.5), (2.4), and (A.1), we see that

$$A(u_h; u_h, v_h) = (f, v_h) = A(u; u, v_h) = A(u; R_h u, v_h),$$

and thus

$$A(u_h; u_h - R_h u, v_h) = A(u; R_h u, v_h) - A(u_h; R_h u, v_h) = T_1 + T_2 + T_3, \tag{A.6}$$

where

$$\begin{aligned} T_1 &= \sum_{K \in \mathcal{T}_h} \int_K (a(u) - a(u_h)) \nabla (R_h u) \cdot \nabla v_h \mathrm{d}x \\ T_2 &= -\sum_{E \in \mathcal{E}_h} \int_E \{(a(u) - a(u_h)) \nabla (R_h u)\} \cdot [v_h] \mathrm{d}s \\ T_3 &= -\sum_{E \in \mathcal{E}_h} \int_E \{(a(u) - a(u_h)) \nabla v_h\} \cdot [R_h u] \mathrm{d}s. \end{aligned} \tag{A.7}$$

To bound T_1, we use Lemma 7 with $z = u_h$ and $w_h = R_h u$, Lemma A.3, and Lemma 3

$$|T_1| \le C |R_h u|_{1,\infty} \|u - u_h\| \, |v_h|_{1,h} \le C(u) h^{r+1} |v_h|_{1,h} \le C(u) h^{r+1} |||v_h|||. \tag{A.8}$$

Using Lemma 8 with $z = u_h$ and $w_h = R_h u$, and applying Lemma 1, Lemma A.3, and Lemma 3 yields

$$\begin{aligned} |T_2| &\le C |R_h u|_{1,\infty} \left(\sum_{K \in \mathcal{T}_h} \sum_{e \in \partial K} h_e \|u - u_h\|_{0,e}^2 \right)^{\frac{1}{2}} |||v_h||| \\ &\le C(u) (\|u - u_h\|^2 + h^2 |u - u_h|_{1,h}^2)^{\frac{1}{2}} |||v_h||| \\ &\le C(u) h^{r+1} |||v_h|||. \end{aligned} \tag{A.9}$$

Applying the fact $[u]_E = 0$ for $E \in \mathcal{E}_h^0$ and $u|_E = 0$ for $E \subset \partial\Omega$, using (2.11), Cauchy-Schwarz inequality, (A.3), Lemma 1, Lemma 3, and (2.1), we have for $h \ll 1$

$$
\begin{aligned}
|T_3| &= \left| \sum_{E\in\mathcal{E}_h} \int_E \{(a(u) - a(u_h))\nabla v_h\} \cdot [u - R_h u] \mathrm{d}s \right| \\
&\le C\|u - R_h u\|_{0,\infty} \left(\sum_{K\in\mathcal{T}_h} \sum_{e\in\partial K} h_e \|u - u_h\|_{0,e}^2 \right)^{\frac{1}{2}} \left(\sum_{K\in\mathcal{T}_h} \sum_{e\in\partial K} h_e^{-1} \|\nabla v_h\|_{0,e}^2 \right)^{\frac{1}{2}} \\
&\le C(u) h^{r+1} |\ln h| \left(\sum_{K\in\mathcal{T}_h} (\|u - u_h\|_{0,K}^2 + h_K^2 |u - u_h|_{1,K}^2) \right)^{\frac{1}{2}} \left(\sum_{K\in\mathcal{T}_h} h_K^{-2} |v_h|_{1,K}^2 \right)^{\frac{1}{2}} \\
&\le C(u) h^{2r+1} |\ln h| \, |||v_h||| \\
&\le C(u) h^{r+1} |||v_h|||. \qquad \text{(A.10)}
\end{aligned}
$$

Now we set $v_h = u_h - R_h u$ in (A.6), use Lemma 6, (A.8), (A.9), and (A.10) to get desired result. □

The following lemma establishes discrete Sobolev inequality in the discontinuous finite element space V_h.

Lemma A.4 *There exists a constant C independent of h and v_h such that*

$$\|v_h\|_{0,\infty} \le C|\ln h|^{\frac{1}{2}} |||v_h|||, \quad \forall v_h \in V_h. \qquad \text{(A.11)}$$

Proof We use (see [6] or Lemma 3.1 in [14])

$$\|v_h\|^2 \le C \sum_{K\in\mathcal{T}_h} |v_h|_{1,K}^2 + C \sum_{E\in\mathcal{E}_h} h_E^{-1} \|[v_h]\|_{0,E}^2, \quad \forall v_h \in V_h, \qquad \text{(A.12)}$$

together with discrete Sobolev inequality [7],

$$\|v_h\|_{0,\infty}^2 \le C(1 + |\ln h|) \left(\sum_{K\in\mathcal{T}_h} \|v_h\|_{1,K}^2 + \sum_{E\in\mathcal{E}_h} h_E^{-1} \|[v_h]\|_{0,E}^2 \right), \quad \forall v_h \in V_h,$$

to get

$$\|v_h\|_{0,\infty} \le C(1 + |\ln h|)^{\frac{1}{2}} \left(\sum_{K\in\mathcal{T}_h} |v_h|_{1,K}^2 + \sum_{E\in\mathcal{E}_h} h_E^{-1} \|[v_h]\|_{0,E}^2 \right)^{\frac{1}{2}} \le C|\ln h|^{\frac{1}{2}} |||v_h|||.$$

□

Now we present the pointwise error estimates of the SIPG method for the quasi-linear elliptic problems in $\mathbb{R}^d$, $d = 2, 3$.

Theorem A.2 *Let $u \in W^{r+1,\infty}(\Omega) \cap H_0^1(\Omega)$ and $u_h \in V_h$ be the solution of (2.4) and (2.5), respectively. Then, for sufficiently small h, there exists a positive constant C independent of*

h such that

$$\|u - u_h\|_{0,\infty} \le C(u)|\ln h|^{\bar{s}} h^{r+1}, \tag{A.13}$$

and

$$\|\nabla(u - u_h)\|_{0,\infty} \le C(u)(h^r + h^{r+1-d/2}), \tag{A.14}$$

where $\bar{s} = \frac{1}{2}$ *if* $r > 1$ *and* $\bar{s} = 1$ *if* $r = 1$.

Proof Using triangle inequality and Lemma A.2, we get

$$\begin{aligned} \|u - u_h\|_{0,\infty} &\le \|u - R_h u\|_{0,\infty} + \|R_h u - u_h\|_{0,\infty} \\ &\le C(u)h^{r+1}|\ln h|^{\bar{r}} + \|R_h u - u_h\|_{0,\infty}. \end{aligned} \tag{A.15}$$

It follows from Lemma A.4 and Theorem A.1 that

$$\|R_h u - u_h\|_{0,\infty} \le C|\ln h|^{\frac{1}{2}} |||R_h u - u_h||| \le C(u)|\ln h|^{\frac{1}{2}} h^{r+1}. \tag{A.16}$$

Combining (A.15) with (A.16) yields

$$\|u - u_h\|_{0,\infty} \le C(u)|\ln h|^{\bar{s}} h^{r+1},$$

where $\bar{s} = 1/2$ if $r > 1$ and $\bar{s} = 1$ if $r = 1$, which completes the proof of (A.13). Using triangle inequality, Lemma A.2, inverse inequality and Theorem A.1, we get

$$\begin{aligned} \|\nabla(u - u_h)\|_{0,\infty} &\le \|\nabla(u - R_h u)\|_{0,\infty} + \|\nabla(R_h u - u_h)\|_{0,\infty} \\ &\le C(u)h^r + Ch^{-d/2}|R_h u - u_h|_{1,h} \\ &\le C(u)h^r + Ch^{-d/2}|||R_h u - u_h||| \\ &\le C(u)(h^r + h^{r+1-d/2}), \end{aligned}$$

which completes the proof of (A.14). □

We use examples in Sect. 4 to give an illustration on these pointwise estimates. We use $\gamma = 10$ in the SIPG. Figures 5 and 6 show the log-log plot of the deviation $(u - u_h)$ against h computed in pointwise sense. As can be seen from the figures, the rates of convergence of the deviation are within the prediction of Theorem A.2.

B. Convergence of Two-Grid SIPG in 3D

To begin with, in establishing the convergence property of the two-grid solution, we need stability estimate $|u_h|_{1,\infty}$. For problems in $\mathbb{R}^3$, this cannot be derived using the procedure in the proof of Lemma 5. Rather, it is proven as a consequence of the pointwise error estimates described earlier. The following lemma formally states this stability.

Lemma B.1 *Let* $u \in W^{2,\infty}(\Omega)$ *and* $u_h \in V_h$ *be the solutions of* (2.4) *and* (2.5), *respectively. Then* $|u_h|_{1,\infty} \le C(u)$.

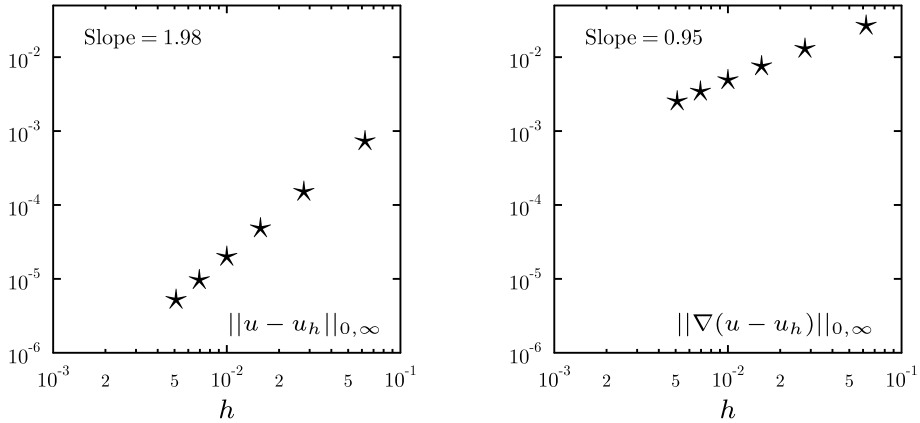

Fig. 5 Pointwise convergence of SIPG solution of (2.2) with elliptic coefficient and true solution as in (4.1)

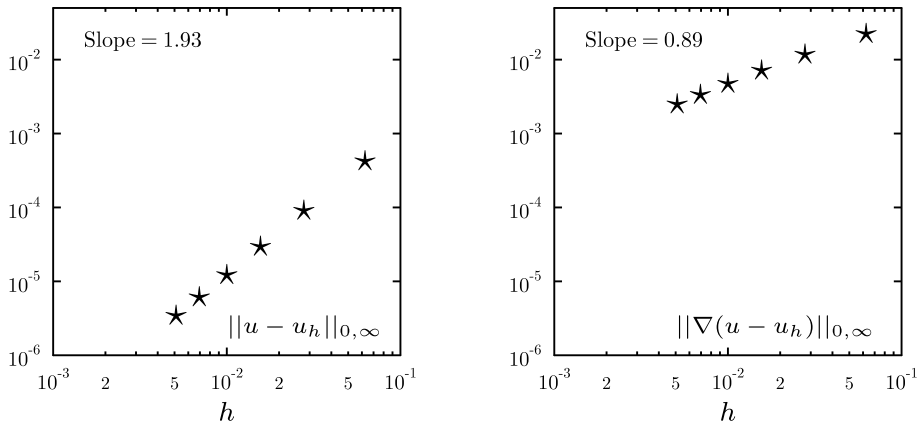

Fig. 6 Pointwise convergence of SIPG solution of (2.2) with elliptic coefficient and true solution as in (4.2)

Proof Using triangle inequality, $|u_h|_{1,\infty} \le |u|_{1,\infty} + |u - u_h|_{1,\infty}$, from which we obtain the desired result by invoking (A.14). □

Finally we state and prove convergence estimates in the broken H^1-norm for the two-grid algorithm of the SIPG method for the quasi-linear elliptic problem in $\Omega \subset \mathbb{R}^3$.

Theorem B.1 *Assume that $u \in W^{r+1,\infty}(\Omega)$, $r \ge 1$, $u_h \in V_h$ and $u^h \in V_h$ are the solutions obtained by* (2.4), (2.5) *and Algorithm* 1, *respectively. For $H \ll 1$, we have*

$$|||u_h - u^h||| \le C(u)H^{r+1}; \qquad |||u - u^h||| \le C(u)(h^r + H^{r+1}). \tag{B.1}$$

Proof The proof is repetition of its counterpart in Theorem 1. In particular, using Lemma B.1, we can reestimate S_1 and S_2. Applying (A.13) to reestimate (3.9), we get

for $|\ln H|^{\bar{s}} H^r \leq 1$

$$|S_3| \leq C(u)\left(|\ln h|^{\bar{s}} h^{2r+1} + |\ln H|^{\bar{s}} H^{2r+1}\right) \leq C(u) H^{r+1}. \tag{B.2}$$

The desired results are obtained by following the steps in the proof of Theorem 1. □

Remark 2 As in the proof of Theorem 1, the proof of Theorem B.1 also requires error estimate of SIPG solution in the L^2-norm (in this case for problems in $\mathbb{R}^3$). This estimate can be derived by using the technique in [22] (see Sect. 6 in [22] for details).

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